

Yupeng Su

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Research Interests

Efficient Pre-training / Post-training Optimization for Large Language Models: focusing on quantization, sparsification, and low-rank decomposition techniques to improve training and inference efficiency.

High-Performance Computing Infrastructure on GPU / FPGA Architectures: including compiler optimization and GPU/FPGA kernel adaptation for high-performance, energy-efficient inference and practical edge deployment.

Education

University of California, Santa Barbara, PhD in Computer Science – CA, USA Sept 2025 – present

- Advisor: Prof. Zheng Zhang (ECE, UCSB).
- GPA: 4.0/4.0.

Southern University of Science and Technology, BE in Microelectronics – China Sept 2021 – July 2025

- Advisor: Prof. Hao Yu (SME, SUSTech).
- GPA: 3.93/4.0, with 178 units completed.
- Awards: Nominee of Top Ten Graduates of SUSTech; Top Ten Graduates of the College of Engineering, SUSTech.
- Dissertation: *Enhanced Mix-Precision Post-Training Quantization for Large Language Models: Block Awareness and Outlier Identification*.

Publications

MuonQ: Enhancing Low-Bit Muon Quantization via Directional Fidelity Optimization July 2026
 Yupeng Su, Ruijie Zhang, Ziyue Liu, Yequan Zhao, Zheng Zhang
arxiv.org/abs/2605.11396 (CoLM, 2026)

RankGuide: Tensor-Rank-Guided Routing and Steering for Efficient Reasoning July 2026
 Jiayi Tian, Yupeng Su, Ryan Solgi, Souvik Kundu, Zheng Zhang
arxiv.org/abs/2604.16694 (CoLM, 2026)

Muon²: Boosting Muon via Adaptive Second-Moment Preconditioning Apr 2026
 Ziyue Liu, Ruijie Zhang, Zhengyang Wang, Yequan Zhao, Yupeng Su, Zi Yang, Zheng Zhang
arxiv.org/abs/2604.09967 (arXiv Preprint, 2026)

MUON+: Towards More Effective Muon via One Additional Normalization Step for LLM Pre-training Feb 2026
 Ruijie Zhang, Yequan Zhao, Ziyue Liu, Zhengyang Wang, Yupeng Su, Liyan Tan, Zheng Zhang
arxiv.org/abs/2602.21545 (arXiv Preprint, 2026)

TEON: Tensorized Orthonormalization Beyond Layer-Wise Muon for Large Language Model Pre-Training Jan 2026
 Ruijie Zhang, Yequan Zhao, Ziyue Liu, Zhengyang Wang, Dongyang Li, Yupeng Su, Sijia Liu, Zheng Zhang
arxiv.org/abs/2601.23261 (arXiv Preprint, 2026)

APTQ+: Attention-FFN-aware Post-Training Quantization for a Layer-wise LLM Accelerator on FPGA May 2026
Ziyi Guan, Zhengfei Chen, Dongkuan Wu, Ao Shen, Yifan Guo, **Yupeng Su**, Graziano Chesi, Mingqiang Huang, Ngai Wong, Hao Yu
[10.1109/TCAD.2026.3697679](#) (IEEE TCAD, 2026)

LLM-Barber: Block-Aware Rebuilder for Sparsity Mask in One-Shot for Large Language Models Nov 2025
Yupeng Su, Ziyi Guan, Xiaoqun Liu, Tianlai Jin, Dongkuan Wu, Chenfei Chen, Graziano Chesi, Ngai Wong, Hao Yu
[10.1109/ICCAD66269.2025.11240704](#) (IEEE/ACM ICCAD, 2025)

PTQTP: Post-Training Quantization to Trit-Planes for Large Language Models Sept 2025
He Xiao, Runming Yang, Qingyao Yang, Wendong Xu, Zhen Li, **Yupeng Su**, Zhengwu Liu, Hongxia Yang, Ngai Wong
[arxiv.org/abs/2509.16989](#) (Under Review)

Quantization Meets Reasoning: Exploring Low-Bit Quantization Degradation for Mathematical Reasoning Jan 2025
Zhen Li*, **Yupeng Su***, Runming Yang, Congkai Xie, Zheng Wang, Zhongwei Xie, Ngai Wong, Hongxia Yang
[arxiv.org/abs/2501.03035](#) (Under Review (*Equal Contribution))

EdgeLLM: A Highly Efficient CPU-FPGA Heterogeneous Edge Accelerator for Large Language Models May 2025
Mingqiang Huang, Ao Shen, Kai Li, Haoxiang Peng, Boyu Li, **Yupeng Su**, Hao Yu
[10.1109/TCSL.2025.3546256](#) (IEEE TCAS I: Regular Papers, 2025)

APTQ: Attention-aware post-training mixed-precision quantization for large language models June 2024
Ziyi Guan, Hantao Huang, **Yupeng Su**, Hong Huang, Ngai Wong, Hao Yu
[10.1145/3649329.3658498](#) (IEEE/ACM DAC, 2024)

Experience

Student Research Assistant, Prof. Hongxia Yang's Lab @ PolyU – Hong Kong, China Nov 2024 – May 2025

- Investigated the effects of compression on reasoning and explored data-driven strategies for restoration.

Student Research Assistant, Next Gen AI (NGai) Lab @ HKU – Hong Kong, China Aug 2024 – Feb 2025

- Contributed to lightweight networks for efficient inference under specified computing architectures.

Summer Intern, High Performance Integrated Circuit Design Lab @ SUSTech – China June 2024 – Sept 2024

- Implemented end-to-end pipelines for compressing, optimizing, and deploying models for efficient devices.
- Published **EdgeLLM** based on the project and its results.